

- 6** When aqueous copper (II) sulfate is mixed with potassium iodide, a white precipitate in a brown solution is formed. The brown solution is decolorised when aqueous sodium thiosulfate is added to it.

What is the formula and the electronic configuration of the copper cation in the white precipitate?

|          | formula of white precipitate | electronic configuration of the copper cation |
|----------|------------------------------|-----------------------------------------------|
| <b>A</b> | $\text{CuI}_2$               | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$               |
| <b>B</b> | $\text{CuI}_2$               | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$          |
| <b>C</b> | $\text{CuI}$                 | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$            |
| <b>D</b> | $\text{CuI}$                 | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^1$          |